

# Apple Revenue Forecasting - Notebook Walkthrough

**File:** AFM244\_S26\_Week11\_001\_Thursday.ipynb **Data:** qSales\_2024.csv

This document walks through the notebook cell by cell. It's a time series regression project that forecasts Apple's quarterly revenue.

## 1. Imports

Loads matplotlib (plotting), statsmodels (regression), numpy, and pandas. Sets pandas to display floats with 2 decimals.

## 2. Load the data

Reads qSales\_2024.csv into a DataFrame called qSales. It has 277 rows covering three companies (Apple, Target, Nintendo), with columns like `datadate` (quarter-end date), `tic` (ticker), and `saleq` (quarterly sales revenue).

## 3. Project roadmap (comment)

Notes the plan: visualize the data first, add a time variable, add dummy variables, then run the regression. States the focus is Apple only.

## 4. Convert dates

Turns the `datadate` column from text into a real datetime type, so it can be plotted and sorted properly.

## 5. Filter to Apple

Creates `apple_sales`, keeping only rows where `tic == 'AAPL'` — 92 quarters from 2001 to 2024.

## 6. Plot the data

Line plot of Apple's revenue (`saleq`) over time (`datadate`), to visually check for trend and seasonality before modeling.

## 7. Add a time variable

Creates a column `time` numbered 1, 2, 3, ... 92 - one integer per quarter, in order. This is the standard approach for time-series regression: instead of using the date directly, you use "time period number" as the independent variable.

## 8. Train/test split

Takes the first 75% of the (chronologically ordered) rows as training data, the last 25% as testing data. Standard for time series - train on the past, test on the most recent periods, rather than a random split.

## 9. Model 1 - simple regression

Fits Ordinary Least Squares (OLS): `saleq ~ time`.

### Result:

```
revenue = -13,536.82 + 1,077.61 * time
```

Revenue trends up by about \$1,077.61 million per quarter on average, across the whole history.

## 10. Prediction intervals

Uses `get_prediction()` to generate an 80% confidence interval around the fitted values - both a “mean” interval (uncertainty in the average) and a wider “obs” interval (uncertainty for an individual observation). Time series forecasts are best given as a range, not a single number.

## 11. Add a seasonal dummy variable

Apple’s fiscal Q1 (the December quarter, which includes the holiday iPhone launch bump) is flagged with `iphone_dv = 1`, all other quarters get 0. Also creates `iphone_interaction = time * iphone_dv`, which lets the trend’s slope differ in that quarter versus others - capturing that the holiday-quarter jump has gotten bigger over time, not just a flat bump.

## 12. Model 2 - regression with dummy + interaction

Fits `saleq ~ time + iphone_dv + iphone_interaction`.

### Result:

```
revenue = -11,044.75 + 933.21 * time - 10,422.13 * iphone_dv + 578.33 * (time * iphone_dv)
```

Normal-quarter growth is about \$933/quarter, but the December quarter has both a level shift and an extra growth rate on top.

## 13. Forecast on the test set

Prepares the test-set inputs (`time`, `iphone_dv`, `iphone_interaction`) for the held-out 25% of quarters, then runs `get_prediction()` at both 80% and 90% confidence to produce forecasted revenue ranges - which can be compared against Apple’s actual reported revenue for those quarters to judge how good the model is.

## Tools used

Python, pandas, numpy, statsmodels, matplotlib

## Note

The CSV also includes Target and Nintendo data, but the notebook filters to Apple only (`tic == 'AAPL'`) right after loading, so those rows are never modeled.